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ABSTRACT

Objective: The primary aim of this study was to assess the efficacy of the weight, urine, thirst (WUT) framework in predicting dehydration after a body water manipulation protocol, while concurrently determining the individual and interactive contributions of the model components.

Methods: The total study sample was 93 participants (female, n = 47), recruited from two institutions. Phase 1 involved collecting daily hydration measures from free-living participants (Study 1, 58 participants for 3 days; Study 2, 35 participants for 7 days). Phase 2 entailed a two-hour passive heating protocol, where participants from Study 2 were randomly assigned to one of three groups that manipulated total body water over 24-hours using passive heating and fluid restriction. During each Phase, participants provided urine samples, underwent body mass measurements, and completed questionnaires pertaining to thirst perception. Morning and 24-hour urine samples were assessed for color, osmolality, and specific gravity. Differences between intervention groups, based on the probability of hydration status, were examined (ANOVA) and ridge regression analysis assessed the relative importance of variables within the WUT model.

Results: The study revealed significant differences among the intervention groups for predicted probability of dehydration, as determined by changes in body mass ($p = 0.001$), urine color ($p = 0.044$), and thirst perception ($p < 0.001$). Binomial ridge regression indicated that change in body mass (58%) and thirst perception (26%) were the most influential predictors of dehydration.

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Conclusions: These data support use of an enhanced version of the WUT model, underscoring the significance of changes in body mass and thirst perception in the assessment of hydration status.

Key Words: WATER, URINE SPECIFIC GRAVITY, BODY MASS, FLUID INTAKE, BODY WATER REGULATION, THIRST PERCEPTION

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INTRODUCTION

It is well established that inadequate fluid intake adversely affects physical and psychological performance on the short-term (i.e., acute dehydration), and may increase the risk of chronic disease long-term (i.e., poor fluid intake behaviors). Thus, efforts to adequately assess hydration status allow for a direct means of addressing changes in total body water. A plethora of methods used to determine or capture changes in hydration status are succinctly summarized in previous literature (1). While laboratory or field-based methods for hydration assessment in research settings provide accurate measures of body water regulation and homeostasis, these measures are not feasible in the general population. Cheuvront and Sawka (2) proposed a framework that includes morning body weight (W), first urine void color (U), and thirst perception (T; WUT framework) as a simple approach for laypersons to determine the adequacy of their fluid intake. Individual changes in W (>1% body mass, kg), U (>5 A.U. using a 1 – 8-point urine color chart (3)), and T (conscious desire to consume fluids) provide evidence of inadequate fluid intake; however, a combination of changes that exceed these thresholds further strengthens the likelihood of insufficient fluid intake.

Despite its simplicity, rigorous science and compelling evidence support the WUT model and the individual variables included in the model. Once body weight decreases by 1% of overall body mass, confidence increases that total body water (TBW) decreased due to deficits in fluid intake, rather than sensible and insensible fluid loss. Similarly, as TBW decreases, plasma osmolality increases, triggering internal mechanisms responsible for the presence of thirst (4), and renal water retention increases in response to increasing arginine vasopressin (AVP) concentrations – thereby concentrating the urine. This increase in urine concentration can be

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accurately matched to a urine color comparison chart and correlates with changes in urine osmolality and urine specific gravity (3).

While the WUT framework is theoretically and scientifically sound, the utility and validity of using this framework has not been fully explored. Sekiguchi and colleagues (5) recently tested the WUT framework and found that exceeding cut-point thresholds on two components (W & U, and U & T) could accurately detect hypohydration, a state of body water deficit, as measured by urine specific gravity and urine osmolality. Work by Keefe et al. (6) and Wardenaar et al. (7) supported and contrasted the work of Sekiguchi et al. (5), and thus the ability of the WUT framework to detect hypohydration, respectively However, neither of these studies tested the WUT framework's ability to detect dehydration following a body water manipulation protocol where body water is actively lost from the body (i.e., sweating), a design that allows for the direct assessment of the efficacy of the proposed model. Importantly, this previous work considered WUT components as binary predictors (i.e., value exceeds a threshold or not), rather than treating each component as a continuous measure that can linearly or non-linearly change with the likelihood of dehydration. Lastly, as evidenced in the work of Sekiguchi et el., Keefe et al., and Wardenaar et al., the WUT framework assumes that each component contributes equally to determining the likelihood of dehydration. Although this conjecture is reasonable based on the originally proposed model, it is unlikely to be accurate, since the sensitivity of each WUT component for predicting dehydration differs in reference to body water regulation, suggesting unequal weighting is likely.

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A properly weighted predictive model using WUT to assess hydration status *en masse* is incredibly appealing due to its simplicity, ease and cost effectiveness. A properly weighted predictive model provides individuals a better understanding as to which of these variables offers an improved ability to determine their fluid needs. However, further development and validation is required to rigorously test the model's predictive accuracy across multiple scenarios. Thus, the purpose of this study was two-fold: 1) to assess the utility of the WUT framework in predicting dehydration following a body water manipulation protocol and 2) to determine each WUT component's independent and interactive importance in predicting dehydration. We hypothesize that changes in each WUT component (body mass loss, increased urine concentration, increased perception of thirst) and the probabilities of hypohydration therein would be significantly different across groups undergoing varying levels of body water manipulation. Further, we hypothesize that changes in body mass would be the component of greatest importance when determining the likelihood of dehydration. The results from this study have broad potential applications in field-based settings for individuals who live, work, and train, in environments likely to induce dehydration and can be used to drive improved hydration behaviors across free-living populations.

METHODS

Study Design

This was approached in two phases; an initial observational design where daily hydration measures were collected in free-living men and women, followed by a randomized design where subjects were allocated into one of three groups that manipulated total body water. The initial phase (Phase 1) of the study combined 24-h and first-morning free-living hydration variables

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across two studies (Study 1 [n = 58], three consecutive days of 24-h free-living hydration assessment; Study 2 [n = 35], seven consecutive days of 24-h free living hydration assessment) to determine daily reference change values for changes in WUT components. The second phase of the study (Phase 2, Study 2 only) consisted of a two-hour passive heating protocol (40°C, 50% relative humidity) followed by randomization to one of three groups that manipulated total body water over a 24-h period (see Measures section for fluid prescription procedures for each group). The study design schematic is illustrated in Figure 1.

Participants

Ninety-three participants (females, n = 47; see Table 1 for other participant demographics) between the ages of 18 – 35 y were recruited from a university in the Southeastern and Northeastern United States. Prior to enrollment, participants completed a general medical health history questionnaire to ensure they did not meet any of the following exclusion criteria: 1) no chronic health problems or diseases that alter body water regulation or one's ability to thermoregulate, 2) no history of exertional heat illness within the last three years and no history of cardiovascular, metabolic or respiratory disease, 3) no previous surgery of the gastrointestinal tract that could impact body water regulation, 4) no pharmacologic drug treatment in the previous 15 days, 5) not exercising more than 10 hours per week, and 6) not actively attempting to gain or lose bodyweight. For female participants, all testing visits occurred during the early follicular phase (days 1-8) of the menstrual cycle to maintain consistency among fluid regulatory hormones across female participants (8). Study 1 tested naturally cycling females (no contraceptive use) only, whereas Study 2 did not control for contraceptive use. However, for Study 2, females were excluded if the contraceptive they used reduced the number

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of menstrual cycles they experienced throughout a 12-month period. All participants provided written and informed consent prior to participation in this study, which was approved by the Institutional Review Boards at the University of North Carolina at Greensboro (#18-0063) and the University of Connecticut (H17-129). All study procedures conformed with the Declaration of Helsinki.

Experimental Protocol

Baseline Testing. Prior to the start of testing, all participants arrived at the laboratory for baseline testing, which included measures of nude body mass (NBM), height, and body fat percentage. Note that throughout this manuscript, SI units of mass (kg) are used to describe weight (W) since electronic scales measure weight as a function of the kilograms of force the mass exerts. Participants were provided two 3L opaque containers, and a 100 mL specimen container and were instructed to collect all urine voided beginning 24 h prior to their first scheduled testing visit. Participants were instructed to use the 100 mL specimen container to provide a first morning urine void that was separate from the 24 h sample.

Experimental Trial. For Phase 1 of this study, a total of 419 d (observations) were included; Study 1 included 174 d (58 participants x 3 consecutive days), and Study 2 included 245 d (35 participants x 7 consecutive days). The number of observations from Study 1 and Study 2 differed due to differences in methodological design to answer those study's respective primary outcome measures. All testing visits took place in the morning (0600-0900 h), with each of the participants' visits taking place within one hour of their first testing visit to minimize circadian rhythm effects on TBW regulation and homeostasis. For each testing visit, participants arrived at

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the laboratory to provide a NBM, return both their 24 h and spot urine samples, and complete questionnaires related to their perceived thirst. Prior to leaving the laboratory, participants were given clean specimen containers for 24 h and spot urine collection for the subsequent testing days. These procedures were replicated for days 2 – 3 (Study 1) and days 2 – 7 (Study 2). Participants were free to go about their normal daily activities and behaviors as the specific goal of this phase was to capture free-living hydration status to improve generalizability of the findings.

On day 7 (Phase 2, Study 2), following the return of the 24 h and first morning void urine samples, a NBM, and rating of perceived thirst, participants entered an environmental chamber and remained in a seated position for two hours (40°C, 50% relative humidity) to induce passive fluid loss. Participants wore shorts, t-shirt, socks, shoes, and their undergarments during this 2 h seated and passive heat exposure. Another NBM measurement was taken at the conclusion of this period to determine acute body water loss. Prior to the post heat exposure NBM, participants toweled themselves dry in order to ensure an accurate measure. At the conclusion of their day 7 lab visit, participants were prescribed a set amount of fluid to consume over the following 24h and were instructed to void all urine produced into the clean containers provided to them. Fluid prescription followed one of three protocols that was assigned using a random number generator. On day 8, participants returned their 24 h urine sample and first morning void urine samples, provided a NBM, and perceived rating of thirst.

Measures

Anthropometric Measures. NBM was measured to the nearest 0.1 kg (Study 1, Defender R7000

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Extreme, OHAUS Corporation, Parsippany, NJ; WB-800S Plus, Tanita Corporation, Tokyo, Japan). Height was measured to the nearest 0.1cm using a wall-mounted stadiometer (Studies 1 and 2, Model 216, Seca, Chino, CA). Body density was calculated using the methods of Jackson and Pollock from the average of two skinfold measures (Lange Skinfold Calipers, Beta Technologies, Santa Cruz, CA) on the right side of the body (9, 10); measures were taken at the chest, abdomen, and thigh for males and suprailiac, triceps, and thigh for women. Body fat percentage was calculated using the Siri equation (11).

Perceived Thirst. Thirst was assessed daily using a 9-point Likert Scale with associated word anchors of 1, "Not Thirsty at All"; 3, "A Little Thirsty"; 5, "Moderately Thirsty"; 7, "Very Thirsty"; and 9, "Very, Very Thirsty" (12).

Urinary Hydration Measures. All urine samples (24-h void and first morning void sample) were measured for urine specific gravity (U_{SG}), urine osmolality (U_{OSMO}) and urine color (U_{COL}). USG was measured via digital refractometry (Reichert TS400). U_{OSMO} was measured in duplicate via freezing point depression (Study 1, Osmo One, Advanced Instruments, Norwood, MA; Study 2, Model 3320, Advanced Instruments, Norwood, MA). U_{COL} was rated on a 8-point widely used and validated color scale (3, 13). 24-h urine volume (U_{VOL}) (Ranger 3000, Ohaus, Parsippany, NJ) was measured to the nearest 0.001g. To obtain accurate 24-h urinary hydration measures, the first morning void sample was combined with the 24-h void following urinalysis of the morning first void sample. Only the first morning void value was used in reference to the WUT model (Table 2).

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Day 8 Fluid Prescription. In Study 2, between day 7 and day 8, participants were randomized to one of 3 different fluid intake conditions: Recommended Intake (Control), Moderate Fluid Restriction (Sweat Loss), and Complete Fluid Restriction (Restriction). Participants in the Control group were provided and told to consume the recommended amount of fluid intake for their sex based on the European Food Safety Authority guidelines (14) (2L for females, 2.5L for males), plus any estimated body water losses incurred from resting in the environmental chamber (rounded to the nearest 500mL) before their arrival to the lab on day 8. The Sweat Loss group was asked to consume only the estimated fluid lost through sweating in the environmental chamber. The Restriction group was asked not to consume any fluids, including any water-heavy foods (high water fruits, vegetables, soups, etc.) prior to their next lab visit.

Statistical Analysis

Step 1: Development of Reference Change Value Non-Linear Equations. Using the free-living data (Study 1 and Study 2), the reference change values (RCV) were calculated for changes in body mass (as percentage change), urine color (as absolute unit change), and thirst (absolute unit change), in agreement with the procedures outlined by Cheuvront et al. (15). Briefly, reference change values utilize serial variation in a measure to determine the degree of change necessary to consider the observed value to be different from regular biological and analytics fluctuation. RCV was determined by Equation 1.

$$RCV = \sqrt{2} \cdot Z \cdot \sqrt{(CV_a{}^2 + CV_i{}^2)}$$

Equation 1. Calculation of Reference Change Values (RCV), where Z refers to the z-score (i.e.,

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number of standard deviations) for a given probability (e.g., z-score of 1.65 equates to a one-tailed probability of 95%), CV_a is analytical variation, and CV_i is biological variation within participants. N.B. In agreement with (16) analytical variation (CV_a) for all measures used in the present study was considered negligible, and thus $CV_a = 0$.

We first fit non-linear models to the association between one-tailed test standard deviations equivalent to a desired probability of 50% (z-score = 0), 80% (z-score = 0.84), 85% (z-score = 1.04), 90% (z-score = 1.28), 95% (z-score = 1.65), and 99% (z-score = 2.33), and the associated RCVs, for all three variables.

Step 2: Applying Reference Change Value Equations to Water Manipulation Intervention. Using only data from Study 2 we computed individual-level dehydration probability on Day 8, following the application of the study intervention, using the non-linear equations developed in Step 1 based on the RCVs. The change in hydration parameters on Day 8 was entered (i.e., *x*) into the equations (seen in Figure 2), and the probability of dehydration (i.e., Y) was determined. The differences between intervention groups for the predicted probability [0 to 1] of dehydration were then assessed via ANOVA. For any significant ANOVA models, Tukey Honest Significant Difference post-hoc tests assessed differences between each group.

Step 3: Assessing the Variable Importance of Body Mass, Urine Color, and Thirst. Because the originally proposed WUT model suggested interaction effects between change in body mass, color, and thirst, as opposed to using each variable independently, we further explored these relationships. Using only data from Day 8 in Study 2, individuals were first categorized as

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Dehydrated (Restriction, Sweat Loss) or *Not Dehydrated* (Control). Next, a binomial ridge regression assessed the prediction of the dehydration category given the main effects of change in body mass, color, and thirst and all possible two-way and three-way interactions between predictors. Ridge regression is similar to generalized linear regression but applies a penalty function to predictors, which is useful for high-dimensional data and when predictors are highly correlated (17). The regression was computed using scaled variables via the *glmnet* package (18). Lastly, in lieu of being able to validly calculate standard errors and thus p-values for regularized regression model coefficients, variable importance (i.e., the relative contribution of each independent variable, each of the three two-way interactions, and the one three-way interaction in predicting the outcome) was extracted from the prediction model to construct a scaled Venn diagram. All statistical analyses were completed using *R* statistical program (code available upon reasonable request). Alpha level was set to $p < 0.05$ for all statistical inference tests.

RESULTS

Daily hydration variables derived from the 24 h measures are provided in Table 2 for Study 1 and 2 and the total sample. The probability of hypohydration given a change in body mass, urine color, and thirst, and the non-linear equation describing this relationship is displayed in Figure 2.

Percent body mass change (mean ± SD%) for the three experimental conditions on Day 8 were -0.0 ± 0.52% (range, -0.82 – 0.48), -1.19 ± 0.89% (range, -2.90 – 0.17), and -1.49 ± 1.11% (range, -2.63 – 0.27) for Control, Sweat Loss, and Restriction conditions, respectively. Further, for Control, Sweat Loss, and Restriction conditions, participants consumed 3190 ± 832 mL

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(range, 2500 – 4500 mL), 984 ± 484 mL (range, 500 – 2000 mL), and 0 mL of fluid between the passive heat exposure and Day 8 measures, respectively. Significant differences ($p = 0.001$) were observed in the predicted probability of dehydration between groups on Day 8 (Figure 3A), given the change in body mass. Post-hoc tests demonstrated that Restriction (Mean ± SD, 0.799 ± 0.23, $p = 0.004$) and Sweat Loss (Mean ± SD, 0.787 ± 0.176, $p = 0.003$) had a significantly greater probability of dehydration compared to Control (Mean ± SD, 0.446 ± 0.234). The mean difference in probabilities between the Restriction and Control conditions was 0.35 [95% CI, 0.11, 0.60], while the mean difference in probabilities between the Sweat Loss and Control conditions was 0.34 [95% CI, -0.11, 0.57]. There was no difference between Restriction and Sweat Loss conditions ($p = 0.991$).

An ANOVA model ($p = 0.044$) demonstrated significant differences in the predicted probability of dehydration between groups on Day 8 (Figure 3B), given the change in urine color. Post-hoc tests demonstrated that Restriction (Mean ± SD, 0.832 ± 0.143, $p = 0.047$) had a significantly greater probability of dehydration compared to Control (Mean ± SD, 0.563 ± 0.343). However, Sweat Loss (Mean ± SD, 0.781 ± 0.137) was not significantly different from Control ($p = 0.111$) or Restriction ($p = 0.869$).

Significant differences ($p < 0.001$) were observed in the predicted probability of dehydration between groups on Day 8 (Figure 3C), given the change in thirst. Post-hoc tests demonstrated that Restriction (Mean ± SD = 0.83 ± 0.152, $p < 0.001$) and Sweat Loss (Mean ± SD = 0.635 ± 0.498, $p < 0.001$) had a significantly greater probability of dehydration compared to Control (Mean ± SD = 0.454 ± 0.328). No difference was observed between Restriction and

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Sweat Loss ($p = 0.539$).

The binomial ridge regression resulted in an $R^2 = 0.83$, indicating that 83% of the variance in the predicted probability of dehydration is explained by W, U, T, and their interactions. The model revealed that change in body mass (relative importance = 58%) and thirst (relative importance = 26%) were the most important variables, while the interaction of change in body mass and color (relative importance = 5%), and change in mass and thirst (relative importance = 5%) were the most important interaction effects. A WUT Venn diagram scaled according to variable importance can be seen in Figure 4.

DISCUSSION

The purpose of this study was to empirically test the WUT framework for the determination of hydration status following a body water manipulation protocol and to evaluate and establish the relative contribution of each independent and interactive component. We hypothesized that the WUT would be an effective framework for predicting a change in hydration status, and that change in nude body mass would be most important in determining hydration status. From these analyses, we have reported empirically based reference change values for each component of the WUT framework, demonstrated the effectiveness of these non-linear models in predicting hydration status following a fluid manipulation trial, and presented an updated WUT model that is scaled based on the independent and combined effects of body mass change, urine color, and thirst. These results support and advance the use of the WUT in assessing hydration status with a simple, easy, and cost-effective method. The reference change values (RCV) presented are based on a large and diverse sample of young adults across more than 400 observations (i.e., days of

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participant data collection), enabling a statistically sound number of observations for the day-to-day variability and increasing the generalizability of these results.

Importantly, RCV equations calculated here are theoretically reasonable; for example, a ~1% change in body mass was determined to be *more likely* or *very likely* dehydrated. Our findings also show that the probability of dehydration was *very likely* (90%) in instances where body mass changes exceeded 1.25% body mass loss, which is largely aligned with previously published findings (16, 19). Specifically, Cheuvront et al. found that a body mass change of 1.3% produced an 80% likelihood of dehydration based on a 5mmol/kg increase in plasma osmolality (16). While the probability of dehydration was lower for similar body mass changes in Cheuvront et al. (16), the approximation of their findings to ours lends to the consistency between samples and allows us to make an informed assumption that the participants in our sample had an acute increase in plasma osmolality. Importantly, the differences in the probability of dehydration between our sample and the calculations by Cheuvront et al. (16) is a result of the lower (approximately 42% lower) day-to-day variability observed in the present study. However, this is expected since the sample size used in the current analysis is more than 5 times larger than Cheuvront et al. (16) thus reducing the impact of more extreme values on average intra-individual variability. It must also be noted that it is commonly thought that the daily upper range for variation in body mass to identify an individual as euhydrated is ± 1% body mass (1, 20, 21), however, based on our findings, this may lend to a *more likely* probability of dehydration, likely reflecting the large inter-individual variability of fluid needs. Yet, this does not change our recommendations from an application perspective: if the day-to-day variation in body mass change exceeds 1% of normal body mass, attempts to consume additional fluids

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throughout the day are warranted.

When examining the probability of dehydration based on urine color and thirst perception, we found that an individual is *very likely* to be dehydrated with a change in urine color that is equivalent to 2 levels on a previously validated scale(3), and a change in thirst of one unit using a 9-point Likert scale. Our findings are consistent with previous literature that show the relationships between dehydration and changes in urinary (13, 22–24) and thirst indices (12, 19, 25). Determining the influence of dehydration on urine color using receiver operator characteristic analyses, McKenzie et al. (13) found that a urine color of 5 A.U. corresponded to a level of dehydration that approximated 2% body mass loss. Cheuvront et al., (16) also observed that a urine color of 6.5 A.U. paralleled 2.5% body mass loss, further supporting the association between dehydration and changes in urine color. Our findings are complementary, in that, a 2 A.U. increase in urine color exhibited a 90% probability of dehydration, and thus could be used independently or in combination with the work by McKenzie and colleagues to assist in guiding fluid intake. For example, following a dehydration protocol, a change in an individual's urine color may change from 3 A.U. (before dehydration protocol) to 5 A.U. (following the dehydration protocol). In this example, the change of 2 A.U. would indicate a 90% probability of dehydration, and by using the findings from McKenzie, et al. (13), may also indicate that the overall magnitude of dehydration would be approximately 2% of body mass loss. While this example is theoretical in nature, it must be acknowledged that by using the results of our findings (i.e., a 2 A.U. change in urine color corresponding to a 90% probability of dehydration) as a standalone may not provide an individual with an indication of how much fluid to consume. We must also consider the findings of previous literature that suggests dietary behaviors may alter

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urine color, thus potentially decreasing the efficacy of this measure as a tool to monitor hydration status (26, 27). Thus, if urine color is to be used as a determination of adequacy of fluid intake, it may be prudent to consider adding in an additional metric such as body mass to be used in combination to determine fluid needs.

Previous work (12, 19, 25, 28, 29) exploring the effect of dehydration on thirst perception has consistently shown that greater levels of dehydration elicit a greater thirst response. However, this previous work did not objectively quantify the likelihood of dehydration based on changes in thirst perception. Our findings add to the existing literature by providing nuanced insights on how changes in first morning thirst can be used to aid in determining the adequacy of fluid intake. Interestingly, we observed that a 1 A.U. change in thirst perception resulted in a 90% probability of dehydration. While this low threshold for increased probability of dehydration with a small change in perceived thirst could be due to the sensitivity of the measure (the Thirst scale used in this study was an ordinal scale that increased by increments of 1 A.U.), the contributions of both volumetric and osmotic factors contributing to the thirst response cannot be neglected (30). We want to highlight that our assessment of first morning perception of thirst is essential as previous literature (28, 29) shows that ingestion of fluids minimizes perceived thirst despite sustained levels of dehydration.

Cheuvront and Sawka (2) proposed the WUT framework for lay use in healthy, active, and low-risk populations as a daily method of determining fluid intake adequacy. While the inception of the framework lacked the prospective validation work to determine its efficacy, the scientific principles guiding its development and use are well established (3, 16, 25, 31). In the

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current study, we implemented a randomized water manipulation protocol to elicit either dehydration or euhydration. Unlike the original WUT framework by Cheuvront and Sawka (2) and the recent validation of the framework by Sekiguchi et al. (5), we scaled the associated Venn diagram based on the data-driven importance of each component. Changes in body mass (relative importance, 58%), was found to be the most important component to use to determine the adequacy of fluid intake, followed by thirst (relative importance, 26%), and then urine color (relative importance, 3%). We also found significant interactions between body mass change and urine color, body mass change and thirst in predicting dehydration, but not between thirst and urine color. Given the differences in weighting of each individual component, as well as the two-way and three-way interactions of each component, the determination of *likely* and *very likely* probably of dehydration proposed in the original model is not equally weighted and a greater emphasis should be placed on day-to-day body mass changes and presence of thirst.

Strengths and Limitations

This study applied rigorous statistical analyses to extensive daily body mass, urinary, and thirst observations from a diverse sample and demonstrated good validity of the WUT against a fluid intake manipulation and allowed us to develop sound reference change values (Step 1 of the analysis) for each component of the WUT framework to use for the subsequent analyses and comparisons. Lastly, the utilization of a randomized process to test the water intake experimental trials adds to the confidence in our findings.

However, there are study limitations that should be noted. First, the data from Study 2 required participants to fully adhere to the fluid restriction protocol. If individuals did not adhere

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to the protocol, we may expect some of the collected variables (e.g., urine osmolality) to reflect this deviation from instructions – providing a reasonable justification for exclusion from analysis. However, because of the correlation between our three WUT variables, any exclusion based on urinary criteria that indicated a lack of strict adherence to the protocol would introduce bias into our results and assume that individual variability in response to these protocols has an upper limit; a claim for which we currently have no basis for making. We, therefore, chose not to exclude any participants from the analysis, yet we could still demonstrate the practicality of the WUT model in most cases. Based on the data presented here, future studies should consider the reasonableness of their data in response to fluid manipulation.

Secondly, and as already alluded, these results should be generalized only to a similar sample of healthy young adults and may not apply to other populations, including those with medical or other thermoregulatory concerns, or those individuals that are regularly exposed to dehydrating activities (e.g., athletes, laborers, soldiers) (32, 33). For example, the RCV models or variable importance values presented here may not apply to older persons, as the thirst mechanism becomes less sensitive with aging (34). The "normal" observed variability in each of these variables may therefore be different (i.e., greater) in other populations, and thus these RCV equations should be recomputed for specific samples of interest. However, the within-person variation in WUT variables is relatively stable, even in disease states, and thus researchers and practitioners should tentatively proceed with applying these results to a wider array of populations. Lastly, the ridge regression and resultant Venn diagram utilized all available data (due to sample size for the fluid manipulation protocol, we did not use training, test, and validation splits) and thus could be overfitting to our specific sample. Caution is therefore

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advised when attempting to apply the relative importance of each WUT component. Although we are confident in the relative ranking of the individual and combined effects of the WUT framework within this study population, the exact contribution to the probability of dehydration is less certain in other populations (e.g., older adults, children, certain illnesses eliciting emesis or diarrhea, etc.). Future investigations could confirm the veracity of these model coefficients and variable importance values.

CONCLUSIONS

Changes in first morning body mass, urine color, and thirst can be used independently, and in combination to determine the adequacy of fluid intake on a day-to-day basis. When using the WUT framework, an approach that can be feasibly performed among a lay population, day-to-day changes in body mass hold the largest weight in determining the likelihood of dehydration, followed by changes in thirst perception, and then urine color. It is necessary to continue exploring the utility of the WUT framework across varying populations such as an ageing and/or diseased population to derive a more comprehensive and valid use case.

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FIGURE LEGENDS

Figure 1. Study 1 and Study 2 sample collection and intervention timeline and resulting analyses.

Figure 2. The probability of dehydration as determined by RCV of a decrease in body mass (A), change in urine color (B), and change in thirst (C).

Figure 3. Differences in the predicted probability of dehydration across study conditions on Day 8, as determined by a decrease in body mass (A), change in urine color (B), and change in thirst (C). * = significantly different from control.

Figure 4. Venn diagram demonstrating the interaction between the change in mass, urine color, and thirst in predicting dehydration on Day 8. Note: The size of each category and the degree of overlap between variables are scaled according to the relative variable importance (%) of the main and interaction effects, respectively, from the ridge regression model.

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Figure 1

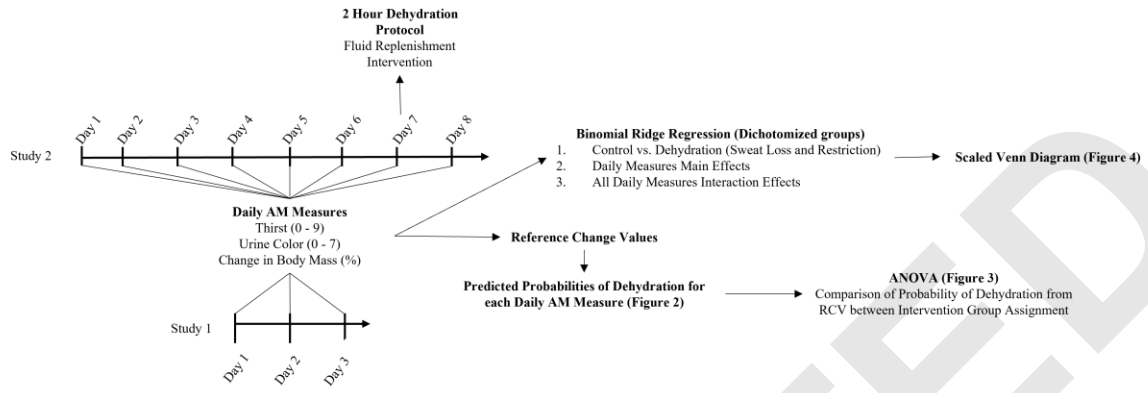

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Figure 2

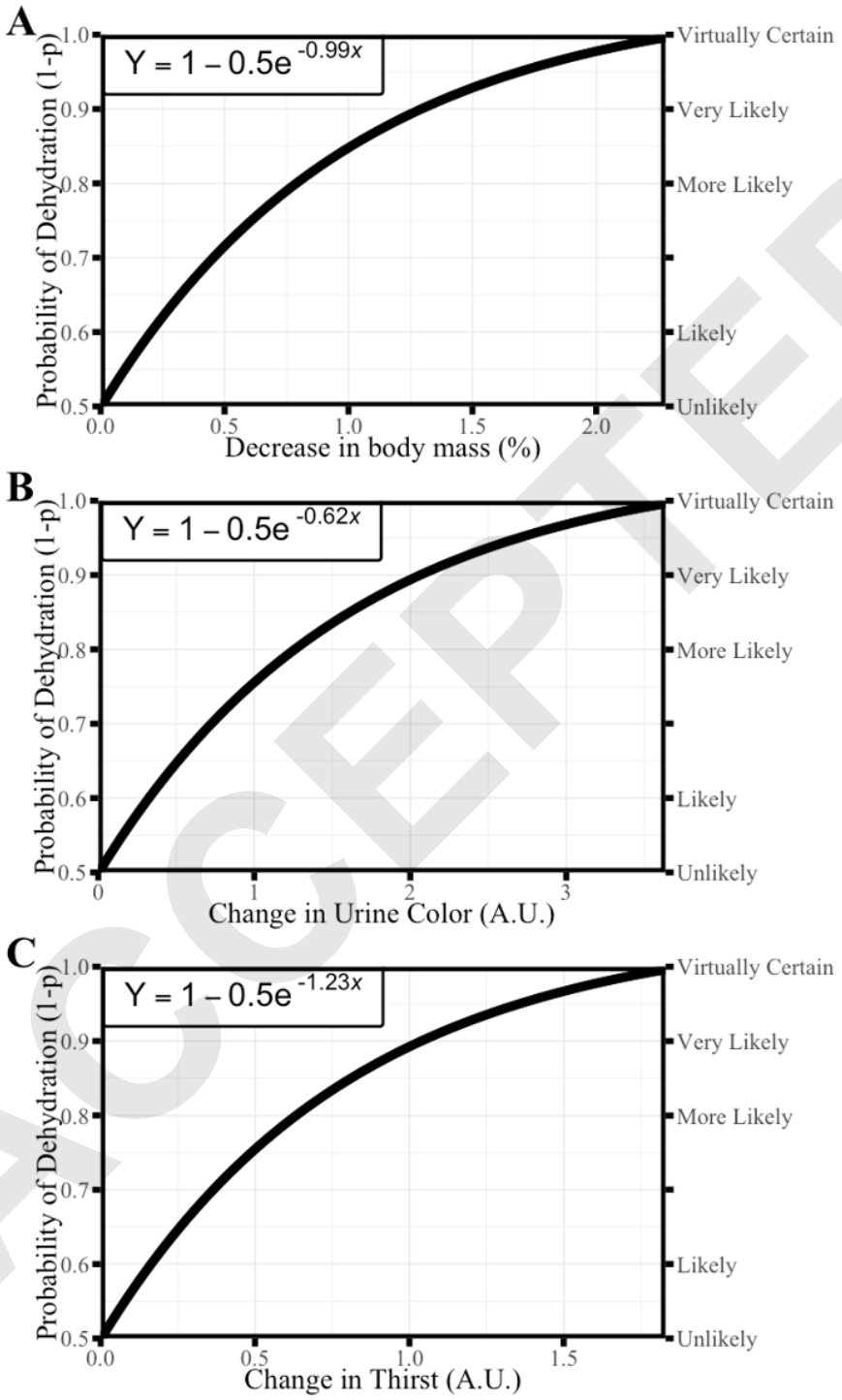

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Figure 3

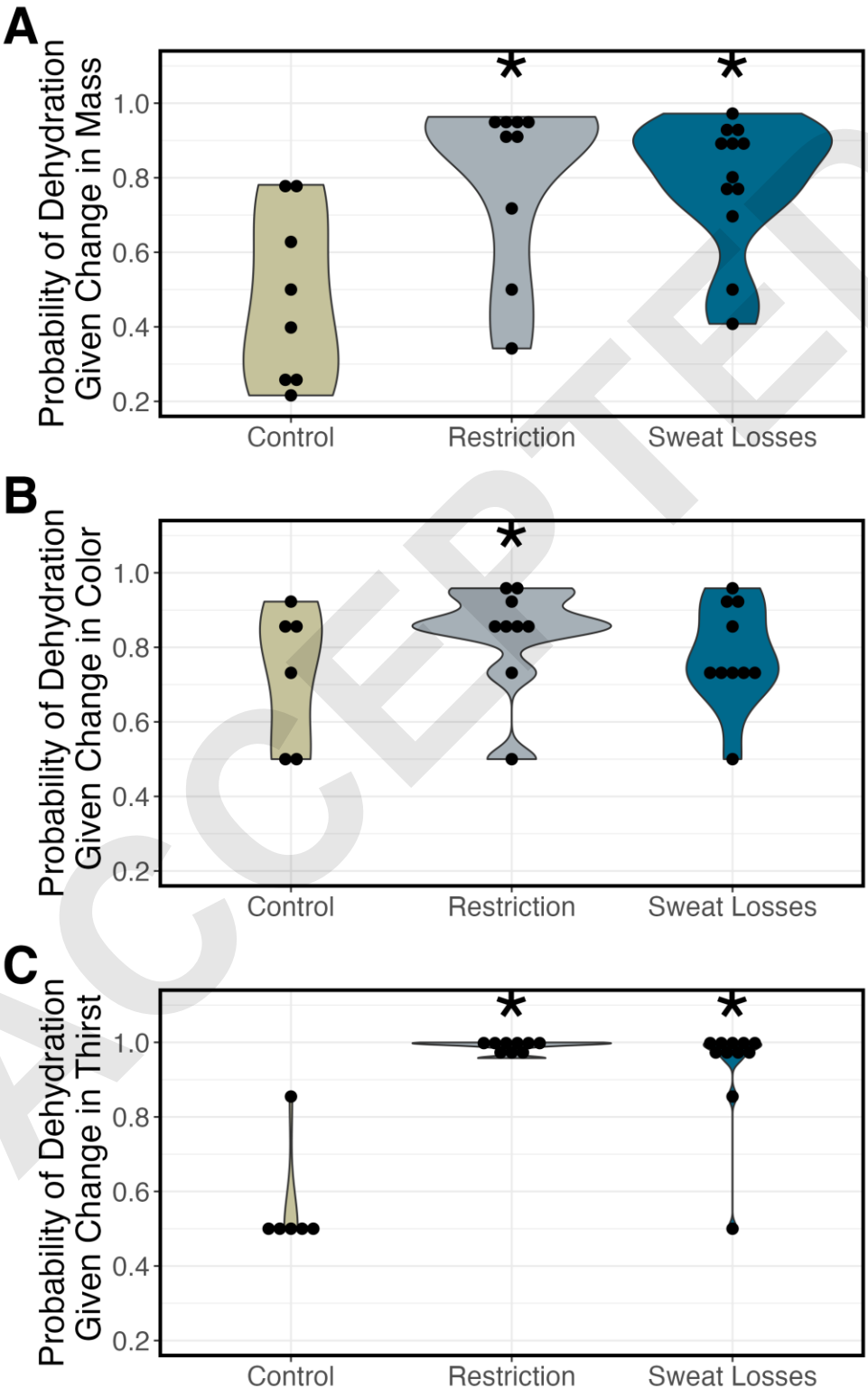

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Figure 4

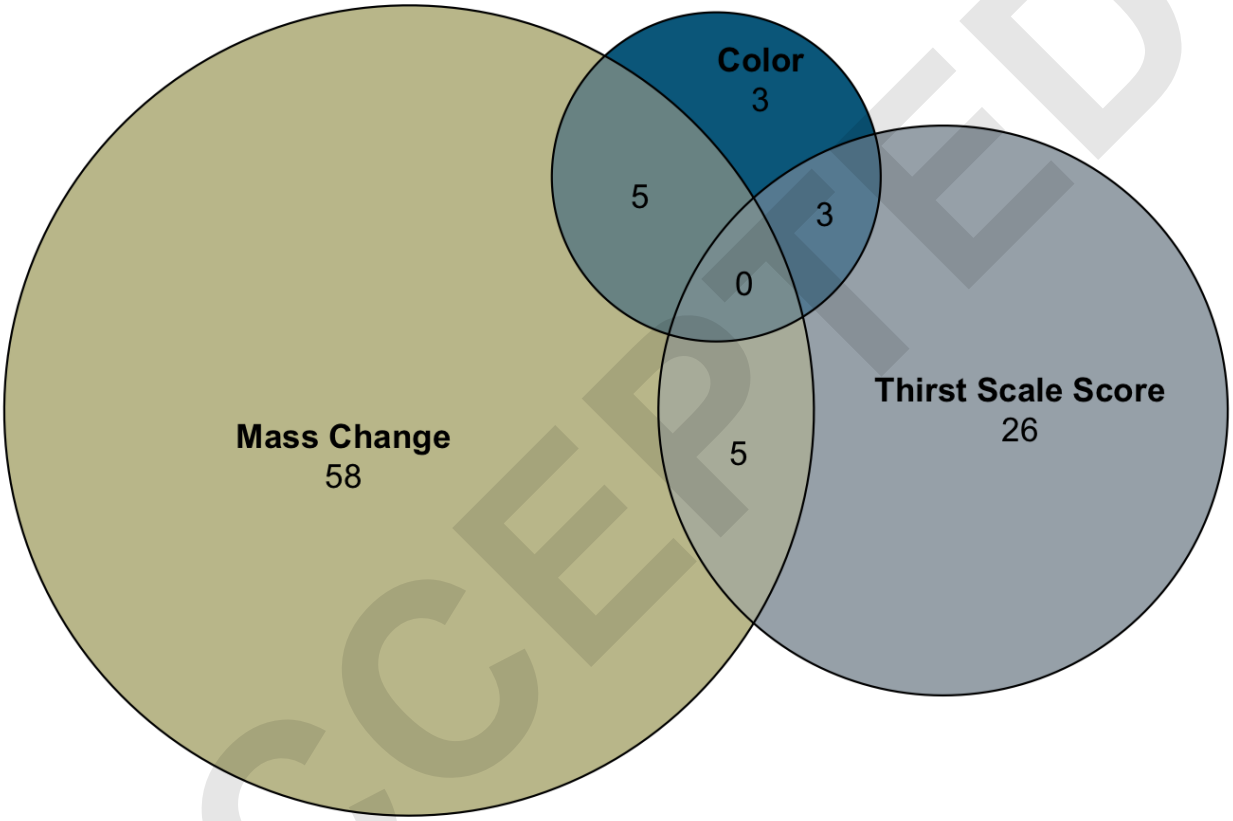

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Table 1. Participant demographics in Study 1 and Study 2

	Study 1	Study 2	Total
Age (y)			
Median [min, max]	20 [18, 25]	23 [18, 32]	20 [18, 32]
Sex			
Female	31	16	47
Male	27	19	46
Ethnicity			
White	NC	19	
Black	NC	16	
Other	NC	0	
Height (cm)			
Mean (SD)	170 (9.8)	172.7 (10.3)	171.3 (10.0
Nude Body Mass (kg)			
Mean (SD)	65.0 (11.9)	77.2 (18.1)	69.8 (15.6)
Body Fat (%)			
Mean (SD)	12.6 (4.3)	18.8 (8.7)	14.9 (6.8)

NC = Data were not collected

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Table 2. Urinalysis values for the 24-hour and first-void morning samples.

		Study 1	Study 2		Total
		Days 1 – 3 Mean (SD), range, CV	Days 1 – 7 Mean (SD), range, CV	Day 8 Mean (SD), range	Mean (SD), range, CV
24 hour	U_{SG}	1.015 (0.009), 1.00 to 1.11, 0.50%	1.016 (0.005), 1.00 to 1.11, 0.46%	1.021 (0.006), 1.01 to 1.03	1.016 (0.008), 1.00 to 1.11, 0.51%
	U_{OSMO}	493 (186), 102 to 1196, 28.4%	567 (210), 164 to 1102, 24.1%	736 (250), 300 to 1142	532 (197), 102 to 1196, 27.8%
	U_{COL}	3.7 (1.0), 1.0 to 8.0, 28.5%	3.9 (1.0), 1.0 to 7.0, 24.7%	4.7 (1.4), 2.0 to 7.0	3.8 (1.0), 1.0 to 8.0, 28.4%
Morning	U_{SG}	1.004 (0.076), 1.013 to 1.03, 4.31%	1.018 (0.005), 1.00 to 1.03, 0.46%	1.025 (0.006), 1.01 to 1.04	1.010 (0.058), 0.013 to 1.04, 2.91%
	U_{OSMO}	629 (188), 108 to 1193, 26.5%	667 (188), 98 to 1182, 26.8%	878 (230), 389 to 1251	655 (183), 98 to 1251, 28.2%
	U_{COL}*	4.6 (1.2), 1.0 to 7.0, 25.5%	4.4 (0.9), 1.0 to 7.0, 26.4%	5.2 (1.4), 3.0 to 7.0	4.5 (1.1), 1.0 to 7.0, 26.6%
	Thirst*	4.4 (1.2), 1.0 to 9.0, 23.6%	3.6 (1.2), 1.0 to 7.0, 32.6%	5.6 (2.3), 1.0 to 9.0	4.1 (1.2), 1.0 to 9.0, 29.9%
	Nude Body Mass (kg)*^	64.64 (10.80), 0.724, 0.76%	77.79 (17.76), 1.44, 0.70%	77.37 (17.42)	70.14 (15.47), 1.05, 0.76%

*Used for all WUT model analyses; ^average difference between minimum and maximum nude body mass (kg) is used instead of the overall range of minimum and maximum nude body mass; CV = mean coefficient of variation (%) of within-participant day-to-day variability.

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